

For #s 1-5, ✓ mean the substitution is a solution to the diff eq.

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Herberty

1. $y' = 4x^3, y = x^4 + 9$

$$(x^4 + 9)' = 4x^3 \quad \leftarrow \text{set equal}$$

$$4x^3 = 4x^3 \quad \checkmark$$

2. $y'' + 25y = 0, y_1 = \cos 5x, y_2 = \sin 5x$

$$y_1' = -5 \sin 5x$$

$$y_2' = 5 \cos 5x$$

$$y_1'' = -25 \cos 5x$$

$$y_2'' = 25 \sin 5x$$

$$-25 \cos 5x + 25 \cos 5x = 0 \quad \checkmark \quad -25 \sin 5x + 25 \sin 5x = 0 \quad \checkmark$$

3. $y' = y + 8e^{-x}, y = 7e^x - 4e^{-x}$

$$(7e^x - 4e^{-x})' = 7e^x - 4e^{-x} + 8e^{-x}$$

$$7e^x + 4e^{-x} = 7e^x + 4e^{-x} \quad \checkmark$$

4. $y'' + y = 35 \cos 6x, y_1 = \cos x - \cos 6x, y_2 = \sin x - \cos 6x$

$$y_1' = -\sin x + 6 \sin 6x$$

$$y_2' = \cos x + 6 \sin 6x$$

$$y_1'' = -\cos x + 36 \cos 6x$$

$$y_2'' = -\sin x + 36 \cos 6x$$

$$-\cancel{\cos x} + 36 \cos 6x + \cancel{\cos x} - \cos 6x = 35 \cos 6x \quad \checkmark$$

$$-\cancel{\sin x} + 36 \cos 6x + \cancel{\sin x} - \cos 6x = 35 \cos 6x \quad \checkmark$$

5. $9x^2 y'' + 135xy' + 441y = 0, y_1 = \frac{1}{x^7}, y_2 = \frac{\ln x}{x^7}$

$$y_1' = \frac{-7}{x^8}$$

$$y_2' = \frac{x^7 \cdot \frac{1}{x} - \ln x \cdot 7x^6}{x^{14}}$$

$$y_1'' = \frac{56}{x^9}$$

$$y_2' = \frac{x^6 (1 - 7 \ln x)}{x^{14}}$$

$$y_2' = \frac{1 - 7 \ln x}{x^8}$$

$$y_2'' = \frac{x^8 \cdot (-\frac{7}{x}) - 8x^7 \cdot (1-7\ln x)}{x^{16}}$$

$$y_2'' = \frac{-7x^7 - 8x^7 + 56x^7 \ln x}{x^{16}}$$

$$= \frac{-15x^7 + 56x^7 \ln x}{x^{16}}$$

$$= \frac{x^7(-15 + 56\ln x)}{x^{16}}$$

$$= \frac{-15 + 56\ln x}{x^9}$$

$$9x^2 \left(\frac{56}{x^9} \right) + 135x \left(\frac{-7}{x^8} \right) + 441 \frac{1}{x^7} = 0$$

$$\frac{504}{x^7} - \frac{945}{x^7} + \frac{441}{x^7} = 0 \quad \checkmark$$

$$9x^2 \cdot \left(\frac{-15 + 56\ln x}{x^9} \right) + 135x \left(\frac{1-7\ln x}{x^8} \right) + 441 \left(\frac{\ln x}{x^7} \right) = 0$$

$$\frac{-135 + 504\ln x}{x^7} + \frac{135 - 945\ln x}{x^7} + \frac{441\ln x}{x^7} = 0 \quad \checkmark$$

6. $5y' = 11y$ $[y = e^{rx} \quad y' = re^{rx}]$

$$5re^{rx} = 11e^{rx} \quad 5r = 11 \quad r = \frac{11}{5}$$

7. $y'' + 2y' - 8y = 0$ $[y = e^{rx} \quad y' = re^{rx} \quad y'' = r^2 e^{rx}]$

$$r^2 e^{rx} + 2re^{rx} - 8e^{rx} = 0$$

$$r^2 + 2r - 8 = 0$$

$$(r+4)(r-2) = 0 \quad r = \{-4, 2\}$$

$$8. y' + 7y = 0 \quad y(x) = Ce^{-7x} \quad y(0) = 2$$

$$y(x) = Ce^{-7x}$$

$$-7Ce^{-7x} + 7Ce^{-7x} = 0 \quad \checkmark$$

$$Ce^0 = 2, \quad C = 2$$

$$9. y' = y + 5 \quad y(x) = Ce^x - 5 \quad y(0) = 7$$

$$y(x) = Ce^x \quad Ce^x = Ce^x - 5 + 5 \quad \checkmark$$

$$Ce^0 - 5 = 7$$

$$C = 12$$

$$10. y' + 12x^2y = 0 \quad y(x) = Ce^{-4x^3} \quad y(0) = 9$$

$$y'(x) = -12x^2 Ce^{-4x^3}$$

$$= -108x^2 e^{-4x^3}$$

$$C = 9$$

$$-108x^2 e^{-4x^3} + 12x^2 (9e^{-4x^3}) = 0$$

$$-108x^2 e^{-4x^3} + 108x^2 e^{-4x^3} = 0 \quad \checkmark \quad 0=0$$

$$11. xy' + 3y = 4x^5 \quad y(x) = \frac{L}{2}x^5 + Cx^{-3} \quad y(2) = 4$$

$$y'(x) = \frac{5x^4}{2} - 3Cx^{-4}$$

$$x \left(\frac{5x^4}{2} - 3Cx^{-4} \right) + 3 \left(\frac{x^5}{2} + Cx^{-3} \right) = 4x^5$$

$$\left(\frac{5x^5}{2} - 3Cx^{-3}\right) + \left(\frac{3x^5}{2} + 3Cx^{-3}\right) = 4x^5$$

$$\frac{8x^5}{2} = 4x^5 \quad \frac{1}{2}(2^5) + C(2^{-3}) = 4$$

$$16 + \frac{C}{8} = 4, \quad C = -96$$

12. $y' = 12x^2 (y^2 + 1)$ $y(x) = \tan(4x^3 + C)$ $y(0) = -1$

$$y'(x) = 12x^2 \cdot \sec^2(4x^3 + C)$$

$$12x^2 \cdot \sec^2(4x^3 + C) = 12x^2 (\tan^2(4x^3 + C) + 1)$$

$$\sec^2(4x^3 + C) = \tan^2(4x^3 + C) + 1$$

$$\tan(4(0)^3 + C) = -1 \quad C = 3\pi/4$$

13. $y' = x + y$

14. $y - y_0 = m(x - x_0)$, $y_0 = 1$, $x_0 = 0$, $m = -1/24/x$

$$y - 1 = -\frac{x}{24/x} \quad \frac{dy}{dx} = -\frac{x}{y-1}$$

15. $\frac{dy}{dx} = \frac{y-x}{x+y}$

16. $\frac{dV}{dt} = KV^2$

17. $\frac{dT}{dt} = K(M-T)$

18/19 - multiple choice

$$20 \text{ a. } \frac{dy}{dt} = -kx^2 \quad y(t) = (C - kt)^{-1}$$

$$y'(t) = -(C - kt)^{-2} \cdot k = \frac{k}{(C - kt)^2}$$

$$\frac{k}{(C - kt)^2} = \frac{k}{(C - kt)^2} \quad \checkmark$$

$$21 \quad P = \frac{1}{(C - kt)} \quad P_0 = 2 \rightarrow C = \frac{1}{2} \quad P = \frac{1}{\frac{1}{2} - kt}$$

$$\frac{dP}{dt} = kP \quad P'(40) = 1, \quad 1 = k(40), \quad k = \frac{1}{1600}$$

$$P(t) = \frac{1}{\frac{1}{2} - \frac{1}{1600}t}, \quad 100 = \frac{1}{\frac{1}{2} - \frac{1}{1600}t} \quad t = 1600 \cdot \frac{49}{100} = 784$$

$$1000 = \frac{1}{\frac{1}{2} - \frac{t}{1600}} - \frac{t}{1600} = \frac{1}{1000} - \frac{560}{1000}$$

$$t = 1600 \cdot \frac{499}{1000}$$

$$22. \text{ a. } xy' = 4y \quad y(x) = cx^4 \quad y'(x) = 4cx^3$$

$$x(4cx^3) = 4(cx^4), \quad 4cx^4 = 4cx^4 \quad \checkmark$$

$$-x^4 = cx^4 \quad \text{where } c = -1$$

$$x^4 = cx^4 \quad \text{where } c = 1$$